Victoria Junior College
JC2 Preliminary Examination 2009
8875 H1 Biology Paper 2

Wednesday **1400 – 1600 h**

16 September 2009 **2 Hours**

Name: ____

CT: ____

Instructions

1. This paper is divided into 2 sections.

   Section A: 4 Structured questions

   Section B: 2 Essay questions

2. **Answer ALL questions in Section A and ONE question in Section B**.

3. For Section A, answers must be written on the spaces provided and nowhere else.
   For Section B, answers must be written on the writing paper.

Question No.	Marks
1	
2	
3	
4	
5 OR 6	Essay
Total	

Section A: Structured questions
Answer all questions

Question 1 (10 marks)

Collagen is the most abundant protein in animals. The basic subunit of collagen is the α-chains which consist predominantly of repeating Gly-X-Y tripeptide motifs. (X and Y represent two different amino acids.) The presence of glycine at every third amino acid is essential to allow the α-chains to adopt the characteristic collagen triple helix. The triple helices can further associate to form fibers with high tensile strength. Collagen molecule is the basic building block of bone and tendon.

(a) Explain the importance of glycine in the formation of collagen triplex helix.

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(b) Describe how association between triple helices allows the formation of fibers with high tensile strength.

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Genetic mutations in the gene that codes for α-chain of collagen results in the formation of weak tendons and fragile bones characteristic of osteogenesis imperfecta or brittle bone disease. These mutations include gene mutations and RNA splicing mutations.

(c) Distinguish between gene mutations and RNA splicing mutations.

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(d) Explain why gene mutations do not always produce mutated collagen whereas mutations involving RNA splicing will always produce mutated collagen.

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Question 2 (10 marks)

Clover is an important crop plant grown as food for sheep and cattle. It is a leguminous plant and its root nodules contain nitrogen-fixing bacteria.
Some clover plants can produce hydrogen cyanide when their tissues are damaged. This is a poisonous compound which will prevent herbivores such as slugs from feeding on the plant. Cyanide is also poisonous to the plants that produce them. Those plants that can produce cyanide are called cyanogenic; those that cannot are called acyanogenic.

Figure 1 below shows how the production of cyanide in a species of European clover is genetically controlled.

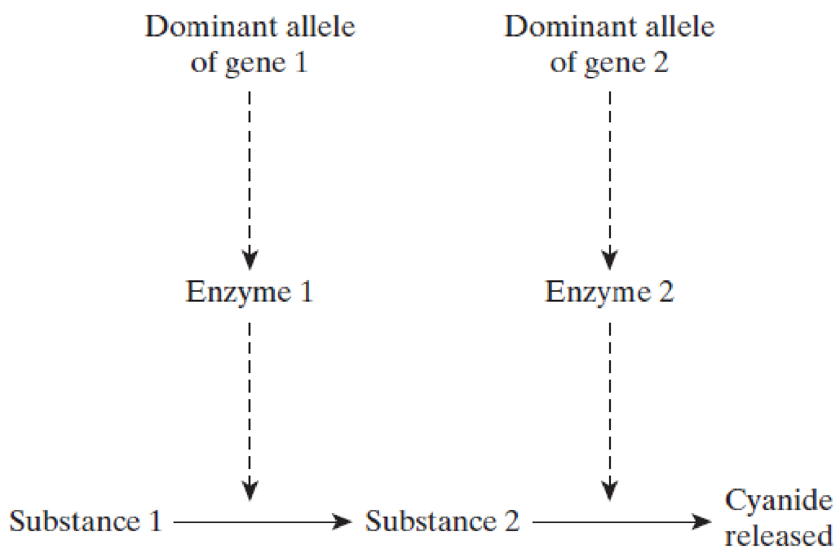

Figure 1

(a) Distinguish between gene and alleles.

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(b) Using appropriate symbols, show by means of a genetic diagram, the different genotypes and phenotypes obtained when two plants that are heterozygous at both the gene loci are crossed. [4]

(c) When the leaves of cyanogenic plants are damaged by slugs, or exposed to low temperatures, membranes within the cells are broken. This causes the release of the enzymes that control the reactions which produce cyanide.

The proportions of cyanogenic and acyanogenic plants in clover populations were determined in different parts of Europe. These are shown in Figure 2, together with the mean minimum winter temperatures. Slugs are not usually active at temperatures below 0°C.

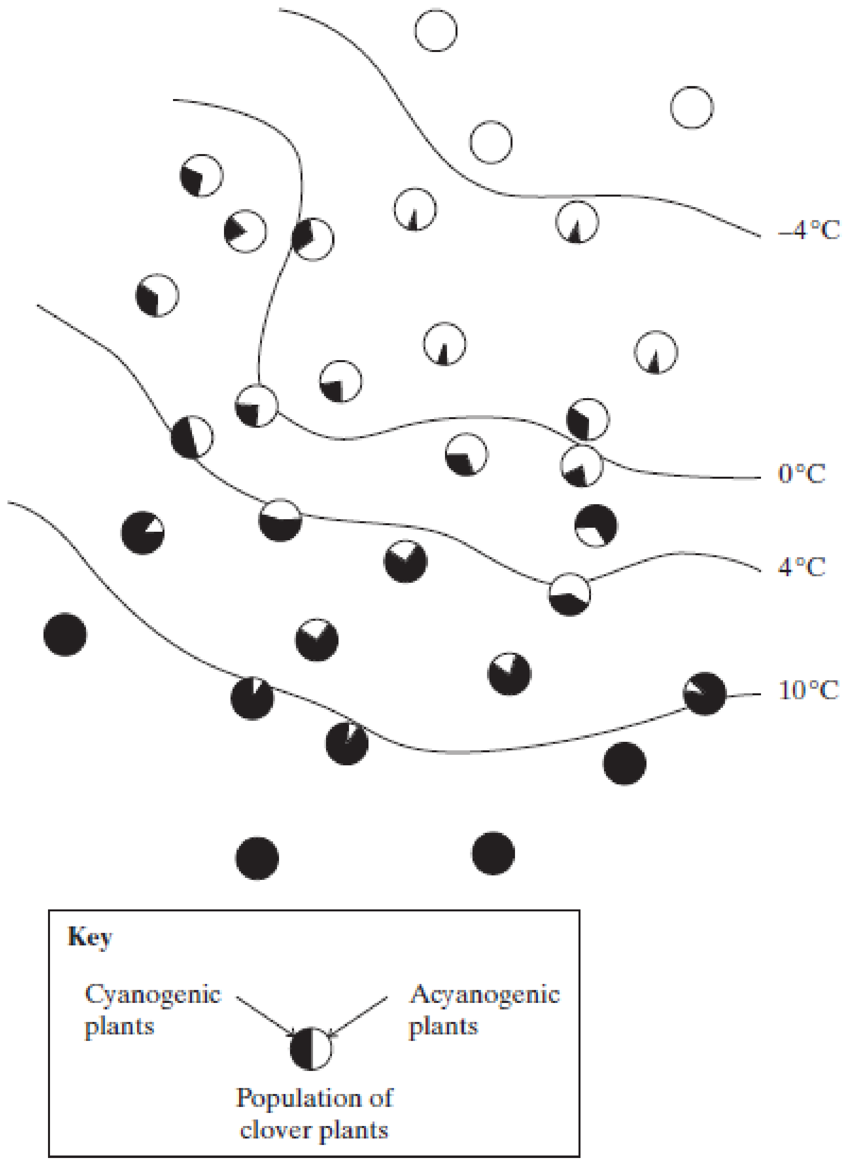

Figure 2

(i) Explain the proportions of cyanogenic and acyanogenic plants in clover populations growing in the area where the mean minimum temperature is below $-4^{o}C$ and in the area where it is above $10^{o}C$.

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Question 3 (10 Marks)

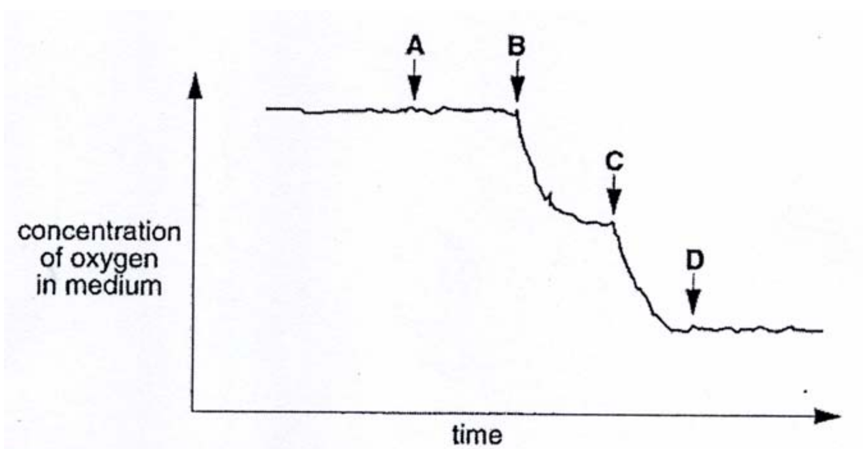
Aerobic respiration of glucose requires enzymes. The process is divided into four stages: glycolysis, the link reaction, Krebs cycle and oxidative phosphorylation.

(a) Complete the table to indicate the precise location of the four stages. [1]

Respiratory Stage	**Location in cell**
Glycolysis	
Link reaction	
Krebs cycle	
Oxidative phosphorylation	

Liver cells are frequently used as a source of mitochondria. These cells are homogenized in a sucrose solution and the mitochondria isolated. Oxygen consumption over time of the suspended mitochondria is then measured and presented in a graph.

The figure below shows the oxygen comsumption of the mitochondria suspension over time. At point **A** the respiratory substrate glucose was added. At points **B**, **C** and **D** equal quantities of ADP were added.



(b) Explain the results

(i) Between points **A** and **B**

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(ii) Between points **B** and **C**

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(iii) After point **D**

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(c) Some of the liver cells were grown in a culture. Radioactive amino acids were added to the solution in which they were being grown. The radioactivity acts as a label on the amino acid so that it can be detected wherever it is. This radioactive label allows amino acids to be followed through the cell. At various times, samples of the cells were taken and the amount of radioactivity in different organelles was measured. The results are shown in the table.

Time after radioactive amino acids were added to the solution/ minutes	**Amount of radioactivity present/ arbitrary units**		
	Golgi apparatus	**Rough endoplasmic reticulum**	**Vesicles**
1	21	120	6
20	42	68	6
40	86	39	8
60	76	28	15
90	50	27	28
120	38	26	56

(c) Using the information in the table, describe a path of radioactivity through and out of the three organelles. Explain what could be happening to the amino acids along the path.

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Question 4 (10 Marks)

Australian scientists have recently produced genetically-engineered clover containing a protein high in sulphur-containing amino acids. As proteins in wool are also rich in these amino acids, it is thought that feeding sheep on this new strain of clover (leaves are consumed) will produce an increase in the yield of wool. A length of DNA was prepared from the sequences in the diagram below for insertion into the clover.

(a) Describe how enzymes are used to construct the DNA fragment shown above.

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(b) Suggest why the following sequences were included in the DNA introduced into the clover.

(i) Sequence B ……………………………………………………………………………

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(ii) Sequence D ……………………………………………………………………………….

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(c) The Ti plasmid in *Agrobacterium* is often used in the genetic engineering of plants. Describe how the Ti plasmid can be used as a vector to introduce the above length of DNA into clover.

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(d) The recombinant Ti plasmid is often reintroduced into *Agrobacterium*. The transformed bacteria is then used to infect plant cells.

Briefly describe two methods which can be used to transfer the recombinant Ti plasmid to *Agrobacterium*.

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Section B
Answer one question only.

Write your answers to this question on the separate answer paper provided. Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be set out in sections (a), (b) etc. as indicated in the question.

5 (a) Using named examples, compare and contrast between storage and structural forms of carbohydrate. [8]

(b) Distinguish between the regulation of protein synthesis in an animal cell with the bacteria cell model. [5]

(c) Explain the roles of nucleic acids in cellular protein synthesis. [7]

6 (a) Compare and contrast between cyclic and non-cyclic photophosphorylation pathways in photosynthesis. [4]

(b) Describe the differences between gene and chromosomal mutation. [6]

(c) Using appropriate examples, describe the properties of stem cells and explain the normal functions of stem cells in a living organism. [10]